Lecture one

Introduction to Information Technology

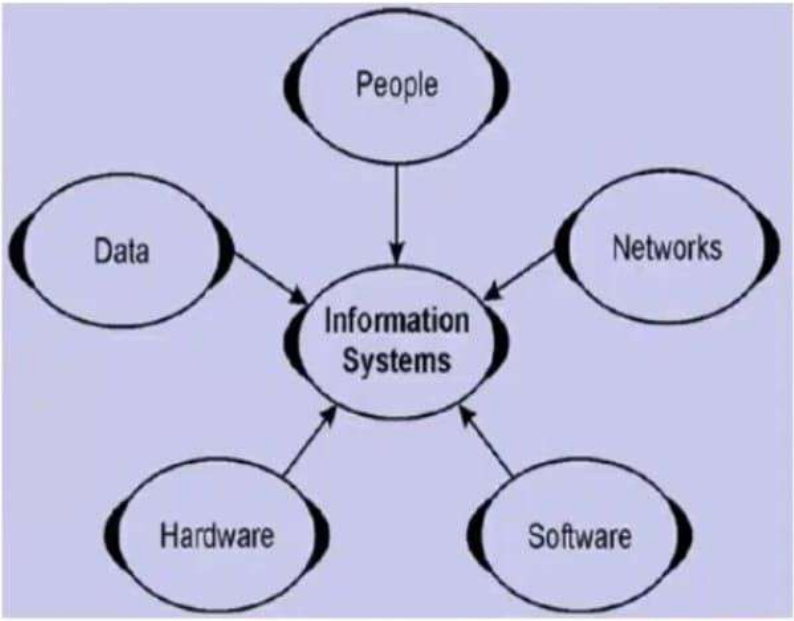
Objective

1. Identify the core components of an information system: people, procedures, software, hardware, data, and the Internet.
2. Contrast system software with application software.
3. Describe the unique roles of the three main types of system software.
4. Compare and contrast general-purpose applications, specialized applications, and mobile apps.
5. List the four main computer categories and specify the five types of personal computers.

6. Detail the various kinds of computer hardware, covering the system unit, input/output devices, storage, and communication equipment.
7. Define "data" and differentiate between document, worksheet, database, and presentation files.
8. Explain key connectivity concepts, such as the wireless revolution, the Internet, cloud computing, and the Internet of Things (IoT).

Information system parts

- People
- Procedures
- Software
- Hardware
- Data
- network



1. People – The Most Important Part

These can be categorized as either **direct**, like drafting documents in a word processor or establishing an internet connection, or **indirect**, where computers serve as essential tools in fields such as healthcare, entertainment, education, and commerce." As follow

- **The Users:** Mr. Karim's waitstaff, his chef (let's call him Hassan) who reads the orders from the kitchen screen, and Mr. Karim himself when he runs the reports on his tablet at home. If they aren't properly trained, don't understand the value of the system, or refuse to use it correctly, the system will fail.

- **The Customers:** They are also part of the system! They might use a tablet on the table to request the bill or pay by credit card. Their interaction generates data.
- **The IT Professionals:** The people who set up the network, install the software, and fix it when it breaks. For a small business like Mr. Karim's, this might be an external consultant like a young engineer named Youssef who comes by once a month.
- **The Management:** Mr. Karim himself, who decides *which* information system to build or buy, and defines the goals it should achieve.

2. Procedures – The Rules and Policies

It are guidelines and policies that govern how the system works and how people should use it.

They are the "operating manual" for the information system. They can be formal written documents or informal, "tribal knowledge" passed between employees. For Mr. Karim's new digital restaurant, procedures might include:

- **For the Waitstaff:** "You must enter the table number *before* you can enter the food order. If a customer pays in cash, you must hit the 'Cash' button and print the receipt for the customer."

- **For the Manager (let's call him Ahmed):** "At the end of each night, you must run the 'End of Day' report from the POS system. Compare the cash in the drawer to the report. Any discrepancy over 50 EGP must be reported to Mr. Karim immediately."
- **For Backup:** "Every Sunday at 3 AM, the system will automatically back up all data. Once a month, the manager must take a backup drive off-site to protect against fire or theft."

3. Software – The Instructions

Hardware is useless without instructions. Software is the set of instructions that tells the hardware what to do. There are two main types:

- **System Software:** This is the behind-the-scenes software that runs the computer itself. The most important example is the operating system, like Windows, macOS, or Linux. It manages the hardware and creates a platform for other software to run on.

- **operating systems:**

Programs that coordinate computer resources, provide an interface between users and the computer, and run applications.

- **Utilities:**
  - Performs tasks related to managing computer resources.
  - Example: antivirus program. These programs protect your computer system from viruses or

malicious programs that are all too often deposited onto your computer from the Internet.

- **Application Software:** This is the software that does the work you want to do. For Mr. Karim, this is where the magic happens.

- **Application software types:**
  - **General-purpose applications** ○ Programs you have to know to be considered an efficient and effective end user.
    - Examples: browsers, Word processor, spreadsheet,..etc.
  - **Specialized applications** ○ Programs focused on specific disciplines and occupations.

- Examples: graphics and Web authoring programs.

- **Mobile apps, known as mobile applications or apps**

4. Hardware The Physical Components

This is the tangible part, the stuff you can kick. In the context of Ali's restaurant, this started with his notepad and pen.

In a modern organization, hardware is much more sophisticated. It includes:

- **Computers:** Desktops, laptops, tablets, and servers (the powerful computers that store and manage data for the whole network).
- **Mobile Devices:** Smartphones and handheld scanners that waiters use to send orders directly to the kitchen.
- **Peripheral Equipment:** Printers for receipts, barcode scanners for inventory, and the screens in the kitchen that display incoming orders.

- **Networking Hardware:** The routers, switches, and cables that connect all the other pieces together, allowing them to communicate.

- **There are four types of computers:**
  - supercomputers
  - mainframe computers
  - midrange computers
  - personal computers
- **Supercomputers**: the most powerful; special highcapacity computers used in very large corporations.
- **Mainframe computers**: not nearly as powerful as supercomputers, they are capable of great processing

speed and data storage; occupy specially wired, air-conditioned rooms.

- **Midrange** computers, also referred to as servers, are computers with processing capabilities less powerful than a mainframe computer yet more powerful than a personal computer.

Personal computers (PCs): least powerful but most widely used and fastest-growing type of computers.

The System Unit: The Powerhouse and Brain

- **Definition:** The metal or plastic case that houses the computer's critical electronic components. It is where the actual processing work gets done.

Key Components Inside the System Unit

1. The Motherboard: The Central Nervous System

- A large, flat circuit board that acts as the main communications hub.
- Every other component plugs into it.
- Contains the wiring (circuits or buses) that allows data and power to travel between all parts.

2. The Central Processing Unit (CPU): The Brain

- Also known as the "processor."
- Executes every instruction from your actions (clicking, typing, opening programs).
- Performs all basic arithmetic, logic, and control operations.
- Acts as the computer's foreman, telling other parts what to do while doing the heavy thinking.

3. Memory (RAM): The Work Desk

- Stands for Random Access Memory.
- A temporary workspace that holds data and instructions for currently running programs.
- **Volatile:** Forgets everything when the power is turned off (hence the need to save your work).

4. Expansion Cards: Adding Specialized Skills

- Circuit boards that plug into slots on the motherboard to add new features or enhance performance.
- **Graphics Card (Video Card):** Handles complex visuals for gaming, 3D design, and video editing.
- **Sound Card:** Enhances audio processing and output quality.
- **Network Card:** Connects to Wi-Fi or wired networks (though often built-in now).

5. Power Supply Unit (PSU): The Heart

- Draws power from the wall outlet and converts it to the correct voltages for all components.
- Distributes electrical power to the motherboard, storage drives, and other parts.
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Summary

- **Input Devices:** Tell the computer what to do.
- **System Unit:** Houses the brain (CPU), workspace (RAM), hub (motherboard), and power (PSU) to process the request.
- **Output Devices:** Display the results to you.
- **Secondary Storage:** Saves your finished work permanently.

5. Data – The Fuel

Hardware and software are just empty vessels. The real value of an information system lies in its **data**. Data are raw facts and figures. On their own, they are meaningless.

- A single sales transaction: "*Table 4, Lasagna, 19.99.*" That's a piece of data.
- An inventory record: "*Parmesan Cheese, 5 units left.*" That's another piece of data.
- A customer's name and phone number for reservations.

Ali's information system takes all these raw data points and turns them into **information**—data that has been processed and organized so it becomes meaningful and useful. For example:

- By collecting data on all sales for a month, Ali can generate a report showing that Friday nights are his busiest time. That's **information**.
- By analyzing sales data, he can see that the Lasagna is his most popular dish and the Calamari isn't selling well. That's **information**, which becomes **knowledge** that helps him decide to put the Calamari on special.

Data flows through the system, gets stored in **databases**, and is the fuel for every decision Ali makes.

6. Networks

A **computer network** is a system that connects two or more computing devices together to share data and resources.
Networks can be classified by size, such as **Local Area Networks (LANs)** which connect devices within a limited area like an office, and **Wide Area Networks (WANs)** which span cities or countries, with the internet being the largest example.
Devices on a network communicate using a set of rules called **protocols**, with the most essential being **TCP/IP** (Transmission Control Protocol/Internet

Protocol), which governs how data is packaged, addressed, and sent across networks.
To connect, devices require both **hardware** like routers, switches, and network interface cards, as well as a unique address known as an **IP address** to identify them on the network.
The primary benefits of networking include **resource sharing** (such as printers and storage), **rapid communication** via email and messaging, and **centralized data management** for easier access and backup.